



Supplementary Materials for

Prevalent, protective, and convergent IgG recognition of SARS-CoV-2 non-RBD spike epitopes

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Other Supplementary Materials for this manuscript include the following:
(available at science.sciencemag.org/cgi/content/full/science.abg5268/DC1)

MDAR Reproducibility Checklist

Materials and Methods

Austin cohort and collection of peripheral blood

All of the SARS-CoV-2 immune plasmas used for this study were collected from non-hospitalized PCR-confirmed individuals who presented with symptomatic disease. Whole blood was collected from convalescent COVID-19 subjects while they were quarantined at home. Blood draws of Study Subjects P1 and P2 occurred at days 12 and 56 post-onset of symptoms, Subject P3 at day 11, and Subject P4 at days 19 and 45. Plasma and PBMCs were separated and collected by density gradient centrifugation using Histopaque-1077 media (Sigma-Aldrich). Informed consent was obtained for all study participants under the University of Texas at Austin IRB protocol #2020-03-0085. Details of sex and age are included in fig. S1.

Expression and purification of SARS-CoV-2 proteins

For Ig-Seq antibody proteomics, the cloning, expression, and purification of the prefusion-stabilized spike ectodomain (S-ECD “S-2P”; GenBank: MN908947) encoding residues 1–1208 and containing two proline substitutions at 986 and 987 as well as other modifications, and residues 319–591 encoding the receptor binding domain (RBD), have been previously described (*1*). For cryo-EM, spike protein was expressed by transiently transfecting plasmid encoding the HexaPro spike variant (*34*) containing substitutions S383C and D985C (*35*) with a C-terminal TwinStrep tag into FreeStyle 293-F cells (Thermo Fisher) using polyethyleneimine, with 5 μ M kifunensine being added 3 hours post-transfection. The cell culture was harvested 4 days after transfection and the medium was separated from the cells by centrifugation. Supernatants were passed through a 0.22- μ m filter followed by passage over StrepTactin resin (IBA). The sample was further purified by size-exclusion chromatography using a Superose 6 10/300 column (GE Healthcare) in buffer containing 2 mM Tris pH 8.0, 200 mM NaCl, and 0.02% NaN₃.

ELISA

The methods for enzyme-linked immunosorbent assay to measure titers of anti-SARS-CoV-2 IgG plasma antibodies have been previously described (*36*). For determination of mAb domain-level reactivity against recombinant spike ECD, RBD, and NTD proteins, a standard indirect ELISA was used. Costar high binding 96-well assay plates (Corning) were coated with antigens (4 μ g/ml) in PBS. Antigens included in-house-produced SARS-COV-2 spike ECD (*1*) (S-ECD), SARS-COV-2 spike RBD, as well as commercially obtained SARS-COV-2 spike NTD (Sino Biological). Antigen-reactive mAbs were detected with goat anti-human IgG (Fab)-horseradish peroxidase (Sigma-Aldrich) conjugate diluted 1:5000 in PBS. After washing with PBST-0.1%, the bound antibody was detected with 3,3',5,5'-tetramethylbenzidine soluble substrate (TMB; Millipore) using a Synergy H1 Microplate Reader (BioTek Instruments, Inc.).

V_H repertoire sequencing

PBMCs were lysed in TRIzol Reagent (Invitrogen) and total RNA was extracted using RNeasy (Qiagen). First strand cDNA was synthesized from 500 ng of mRNA using SuperScript IV (Invitrogen), and cDNA encoding the V_H regions of the IgG, IgA, and IgM repertoires was amplified with a multiplex primer set (*37*) using the FastStart High Fidelity PCR System (Roche) under the following conditions: 2 min at 95°C; four cycles of 92°C for 30 s, 50°C for 30 s, 72°C for 1 min; four cycles of 92°C for 30 s, 55°C for 30 s, 72°C for 1 min; 22 cycles of 92°C for 30 s, 63°C for 30 s, 72°C for 1 min; 72°C for 7 min; hold at 4°C, as previously described (*37*). Products were sequenced by 2×300 paired-end Illumina MiSeq.

Paired V_H:V_L repertoire sequencing

PBMCs were co-emulsified with oligo d(T)25 magnetic beads (New England Biolabs) in lysis buffer (100 mM Tris pH 7.5, 500 mM LiCl, 10 mM EDTA, 1% lithium dodecyl sulfate, and 5 mM dithiothreitol) using a custom flow-focusing device as previously described (38). The magnetic beads were washed, resuspended in a one-step RT-PCR solution with an overlap extension V_H and V_L primer set as previously described (38), emulsified using a dispersion tube (IKA), and subjected to overlap-extension RT-PCR under the following conditions: 30 min at 55°C followed by 2 min at 94°C; four cycles of 94°C for 30 s, 50°C for 30 s, 72°C for 2 min; four cycles of 94°C for 30 s, 55°C for 30 s, 72°C for 2 min; 32 cycles of 94°C for 30 s, 60°C for 30 s, 72°C for 2 min; 72°C for 7 min; hold at 4°C. Amplicons were extracted from the emulsions, further amplified using a nested PCR, and sequenced using 2×300 paired-end Illumina MiSeq.

Ig-Seq sample preparation and mass spectrometry

Total IgG was isolated from 1 ml plasma using Protein G Plus Agarose (Pierce Thermo Fisher Scientific) affinity chromatography and cleaved into F(ab')₂ fragments using IdeS. SARS-COV-2 Spike-specific F(ab')₂ was isolated by affinity chromatography using recombinant antigen (1 mg SARS-CoV-2 S-2P or RBD) coupled to 0.05 mg of dry NHS-activated agarose resin (Thermo Fisher Scientific) as follows. F(ab')₂ (10 mg/ml in PBS) was rotated with antigen-conjugated affinity resin for 1 hour, loaded into 0.5-ml spin columns, washed 12X with 0.4 ml of Dulbecco's PBS, and eluted with 0.5-mL fractions of 1% formic acid. IgG-containing elution fractions were concentrated to dryness in a speed-vac, resuspended in ddH₂O, combined, neutralized with 1 M Tris/3 M NaOH, and prepared for liquid chromatography–tandem mass spectrometry (LC-MS/MS) as described previously (13) with the modifications that (i) peptide separation using acetonitrile gradient was run for 120 min and (ii) data were collected on an Orbitrap Fusion (Thermo Fisher Scientific) operated at 120,000 resolution using HCD (higher-energy collisional dissociation) in topspeed mode with a 3 s cycle time.

Bioinformatic analysis

Raw Illumina MiSeq output sequences were trimmed according to sequence quality using Trimmomatic (39) and annotated using MiXCR (40). Sequences with ≥2 reads were clustered into clonal lineages defined by 90% CDR-H3 amino acid identity using USEARCH (41). LC-MS/MS search databases were prepared as previously described (13), using custom Python scripts (available upon request). MS searches, and MS data analyses were performed as previously described (13), adjusting the stringency of the elution XIC:flowthrough XIC filter to 2:1. Final diversity analyses used a lineage abundance cutoff of 0.5% of total CDR3 XIC abundance in the elution. Multiple alignment of IGHV1-24 VH genes (Fig. 3C) was performed using ClustalW at the ExPASy website (<https://embnet.vital-it.ch/software/ClustalW.html>) with scoring matrix equal to identity, opening and end gap penalties set to one, and ending gap and separation gap penalty = 0.05. Amino acid positions of the multiply aligned sequences (Fig. 3C and Table S4) were assigned according to the IMGT numbering system (42).

Antibody expression and purification

Cognate V_H and V_L antibody sequences of interest were ordered as gBlocks (Integrated DNA Technologies) and cloned into a customized pcDNA 3.4 vector containing a human IgG1 Fc region. V_H and V_L plasmids were mixed at 1:2 ratio and were transfected into Expi293F cells

(Thermo Fisher Scientific), which were cultured at 37°C and 8% CO₂ for 5 days, then neutralized and centrifuged at 1000g for 10 min. Antibodies were isolated from filtered supernatants using Protein G Plus Agarose (Pierce Thermo Fisher Scientific) affinity chromatography, washed with 20 column volumes of PBS, eluted with 100 mM glycine-HCl pH 2.5, and neutralized with 1 M Tris-HCl pH 8.0. The antibodies were buffer-exchanged into PBS and concentrated using 10,000 MWCO Vivaspin centrifugal spin columns (Sartorius).

Binding affinity and checkerboard competition by biolayer interferometry

Bio-Layer interferometry (BLI) assays were performed using an eight-channel Octet RED96e instrument (ForteBio). Anti-hIgG Fc Capture (AHC) Biosensors (ForteBio 18-5060) were used and the assay was performed at 25°C with shaking at 1,000 rpm. For IgG1 mAb K_D measurement, antibodies were diluted to 7.5 µg/ml and immobilized onto biosensors. The serially diluted HexaPro (34) (200 nM to 12.5 nM) was associated for 3 min and dissociated for 5 min. The K_D were calculated using a 1:1 binding with drifting baseline model in BIAevaluation software as described (43). Determination of NTD-binding epitopes (checkerboard experiment) was performed using Anti-hIgG Fc Capture (AHC) Biosensors (ForteBio Inc., 18-5060) at 25°C with shaking at 1,000 rpm. The first antibody was captured at 40 µg/ml for 10 min and blocked with 50 µg/ml of IgG isotype control for 5 min. Antigen (NTD, 100 µg/ml) was associated for 5 min and 40 µg/ml of second antibody was associated for 5 min. The ForteBio Octet Data Analysis Software 9.0 was used for all analyses.

For production of spike variants used in the BLI assays (Fig. 4E), mutations were introduced into a HexaPro mammalian cell surface display plasmid as previously described (44). Briefly, the spike coding region was divided into five parts and cloned into a pcDNA5-based vector along with a 3X-FLAG, a Strep Tag II epitope, an HRV 3C protease site, and a PDGFR-β transmembrane domain using a high throughput automated Golden Gate cloning pipeline. HEK293T cells were transfected with the spike display variant plasmids, incubated for 48 hours, washed with PBS, and resuspended in 3C cleavage buffer (150 mM NaCl, 50 mM Tris-HCl pH 8.0). Spike variants were cleaved from the cell surface using 3C protease (Thermo Fisher; 88946), supernatant containing the spike protein was collected by centrifugation at 16,000g for 1 min, and spike was purified through a StrepTactin resin (IBA) column followed by size-exclusion chromatography using a Superose 6 Increase 10/300 column (GE Healthcare). Anti-mouse Fc capture (AMC) Biosensors (ForteBio 18-5089) were hydrated with BLI buffer for 10 min and mouse anti-FLAG IgG was immobilized to the AMC sensor tip. Spike variants were diluted twofold with BLI buffer prior to loading onto the biosensor for binding analysis.

SARS-CoV-2 Microneutralization Assay

USAMRIID: ATCC Vero E6 cells were seeded on 96-well plates 24 hours prior to infection. MAbs were normalized, threefold serially diluted, and incubated with a pre-titrated amount of SARS-CoV-2 virus (SARS-CoV-2/MT020880.1 isolate) at 37°C for 1 hour. The virus-antibody inoculum was added to the Vero E6 monolayers and incubated for 24 hours. Cells were then formalin fixed, permeabilized, and stained with a SARS-CoV nucleocapsid-specific antibody. After counterstaining, the monolayer was imaged under immunofluorescence software analyzed to quantify the presence of the detected antigen.

UNC: SARS-CoV-2 WA1 molecular clone (GenBank accession MT461669) and nLuc virus were generated previously (45). ATCC Vero E6 cells were seeded at 20,000 cells per well on 96-well plates prior to infection. MAb samples were tested at starting concentrations of 30–0.1 µg/ml

and were serially diluted threefold up to eight dilution spots. Diluted mAbs were mixed with 87 PFU/well WT-nLuc virus, incubated at 37°C with 5% CO₂ for 1 hour. Following incubation, the growth medium was then removed and virus–antibody mixtures were added to the cells in duplicate. Virus-only controls were included in each plate. After a 48-hour incubation at 37°C with 5% CO₂, cells were lysed, and luciferase activity was measured using Nano-Glo Luciferase Assay System (Promega) according to the manufacturer’s specifications. Neutralization titers were defined as the sample dilution at which a 50% reduction in relative light units (RLU) was observed relative to the average of the virus-only control wells.

Evaluation of mAb prophylactic efficacy in the MA10 mouse model

For the inhibition of SARS-CoV-2 in the standard laboratory BALB/c mouse model, a pathogenic mouse ACE2-adapted SARS-CoV-2 variant, MA10, was constructed previously (23, 45). At 12-hour before infection, twelve-month-old female BALB/c mice (n=5/group) were injected intraperitoneally with 200 µg/mouse of mAb or PBS. The mice were infected intranasally with a lethal dose (10⁵ or 10⁴ PFU) of the MA10 virus. Body weight of individual mice was measured daily, and all the mice were euthanized at day 4 post-infection by isoflurane overdose. The right caudal lung lobe was harvested and preserved in PBS at –80°C. Viral titers in the lung tissue were measured by plaque assay on Vero E6 cells. The UNC Chapel Hill IACUC protocol for the mouse research is 20-114.

EM sample prep and data collection

Purified spike (>98% purity by SDS PAGE) and Fab CM25 were combined at a final concentration of 0.2 mg/ml and 0.8 mg/ml respectively in buffer containing 2 mM Tris-Cl pH 8.0, 200 mM NaCl, and 0.02% NaN₃. Following a 30-min incubation on ice, 3 µL of the complex was deposited on Au-300 1.2/1.3 grids (UltrAuFoil) that had been plasma cleaned for 4 min in a Solarus 950 plasma cleaner (Gatan) with a 4:1 ratio of O₂/H₂. The excess liquid was blotted for 3 s with a force of –4 using a Vitrobot Mark IV (Thermo Fisher) and plunge frozen into liquid ethane. A total of 3,339 micrographs were collected from a single grid with the stage at a 30° tilt using a Titan Krios (Thermo Fisher) equipped with a K3 detector (Gatan). Movies were collected using SerialEM at 22,500X magnification with a corresponding calibrated pixel size of 1.1 Å² per pixel. A full description of the data collection parameters can be found in Table S5.

Cryogenic electron microscopy (cryo-EM)

Motion correction was performed in Warp (46). Micrographs were then imported into cryoSPARC v2.15.0 for CTF-estimation, particle picking, 2D classification, ab initio reconstruction, heterogeneous 3D refinement and homogeneous refinement (47). 2D classification was performed using 80 classes and an uncertainty factor of 1, followed by a second round of 2D classification using 100 classes and an uncertainty factor of 3. Selected particles were then input into an ab initio reconstruction job with three classes followed by heterogeneous refinement of those three classes. The highest quality class underwent homogeneous refinement with C3 symmetry applied. To improve the CM25–NTD interface density, C3 symmetry expansion was performed followed by particle subtraction using a mask created in UCSF ChimeraX (48) that encompassed two of the three CM25 Fabs and all of the spike density except for the NTD bound to the third CM25 Fab. Local refinement (cryoSPARC v3.1.0) was performed using a mask encompassing one NTD and the CM25 Fab bound to it. Local refinement was limited to 3 iterations and performed using fulcrum coordinates (x,y,z) of 235.7, 228.9, and 210.6. The local shift and

rotation searches were limited to 2 pixels and 6 degrees, respectively. To improve map quality, the focused refinement volumes were processed using the DeepEMhancer tool via COSMIC² science gateway (49). An initial model was generated by docking PDBID: 6VSB (1) and a Fab model based on the CM25 sequence built using SAbPred ABodyBuilder (50) into map density via UCSF Chimera (51). The model was built further and iteratively refined using a combination of Coot, Phenix, and ISOLDE (52-54). The full cryo-EM processing workflow and structure validation can be found in Figs. S7 and S8.

Statistics

GraphPad Prism version 9.0.0 (GraphPad Software Inc., La Jolla, CA, USA) was used to conduct statistical analyses. Non-parametric Mann–Whitney *U* tests and analysis of variance on ranks (Kruskal–Wallis *H* tests) were used to determine the statistical significance of population means between two or more groups, respectively. Statistical differences in MA10 mouse modeling were tested using a one-way ANOVA with Dunnet’s multiple comparisons test, comparing every group with the mock-challenge lung titers.

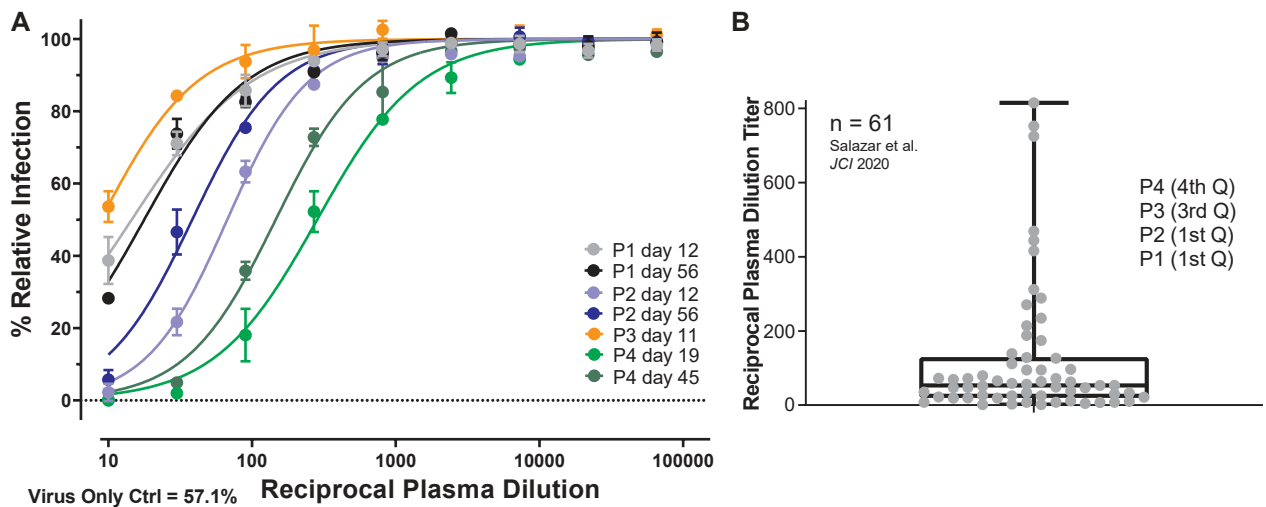


Fig. S1. Live virus neutralization titers of four COVID+ study subjects' plasma at each collection time point.

(A) Serial dilutions of plasma were tested in duplicate (SD error bars) for inhibition of live SARS-CoV-2 virus infection of in vitro monolayered Vero E6 cells. The percent of infected Vero E6 cells in each sample dilution was normalized relative to the virus-only (no plasma) negative control sample. (B) Study subjects examined in this work have a distribution of plasma neutralization titers that span from the 1st quartile (Q) to the 4th quartile as compared to a previously examined cohort of study subjects in Salazar et al. JCI 2020.

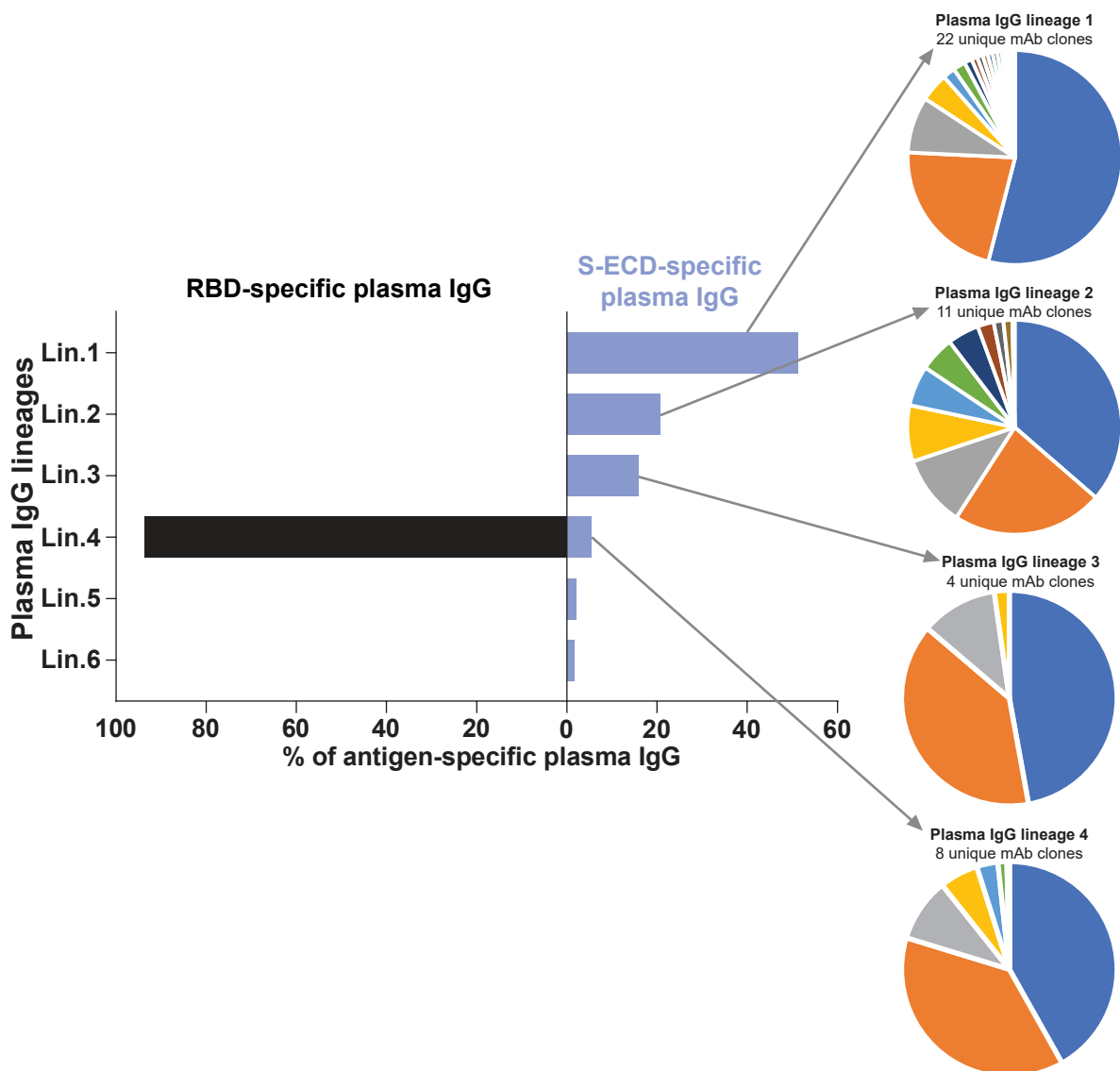


Fig. S2. Ig-seq intralineage diversification in study subject P3 at day 11.

The top four plasma IgG lineages from subject P3 demonstrate a large number of LC-MS/MS identified unique CDR-H3 clones within each lineage (45 total unique CDR-H3 clones in top four IgG lineages combined). This indicates extensive ongoing diversification within this donor at early convalescence.

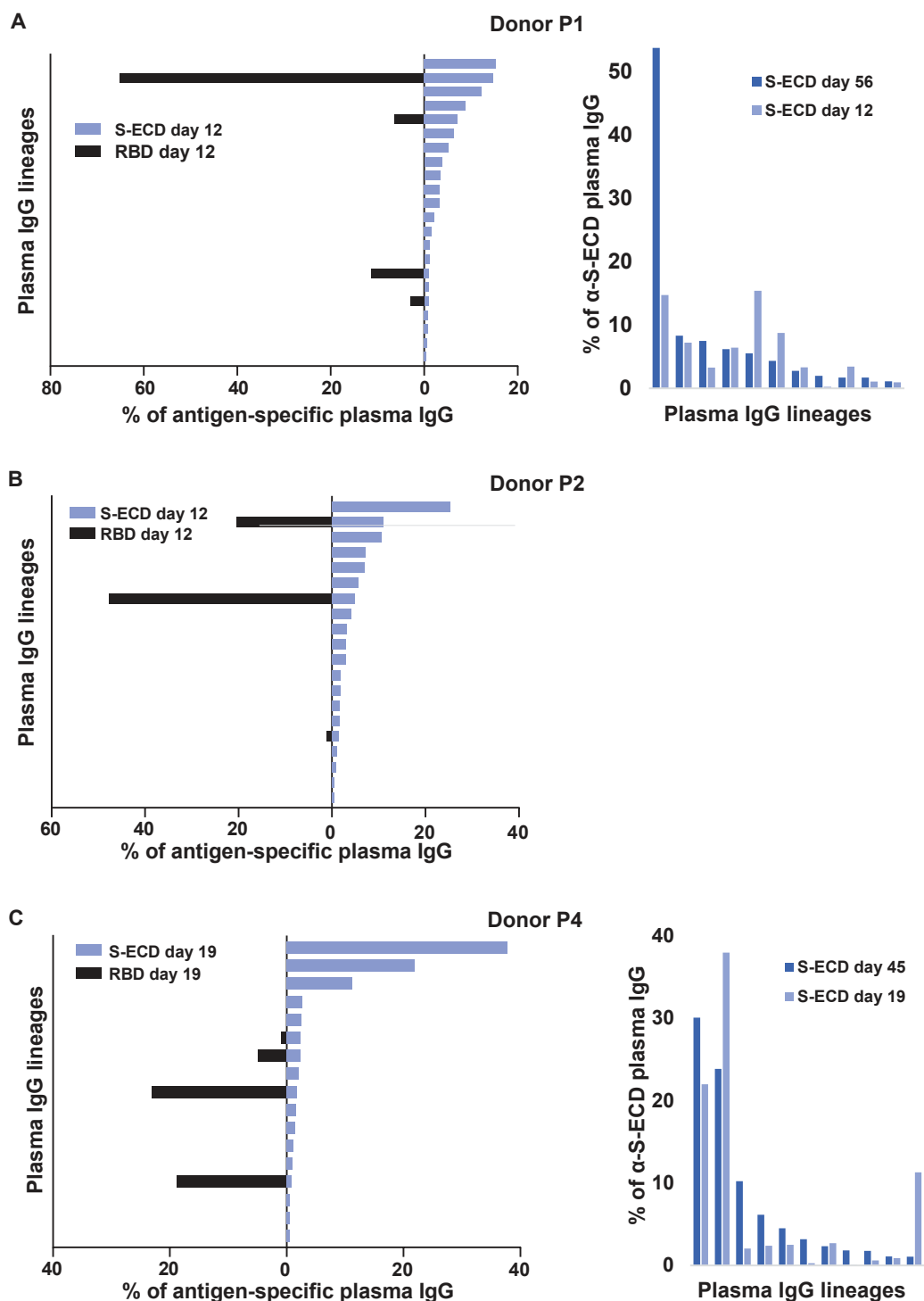


Fig. S3. Ig-seq plasma IgG lineage profiles of study subjects at early and late convalescent time points.

On the left, the first time point Ig-seq profile (days 11-19) for each subject (subject P3 found in Fig.1) shows both the SARS-CoV-2 spike ECD (S-ECD) and RBD abundance for each lineage detected at >0.5% S-ECD plasma IgG (summed lineage XIC). Similarly, on the right, the second time point data for S-ECD (days 45-56) is shown for each lineage detected at >0.5% S-ECD plasma IgG abundance (time point 2), alongside early time point S-ECD data for comparison (subject P2 found in Fig.2).

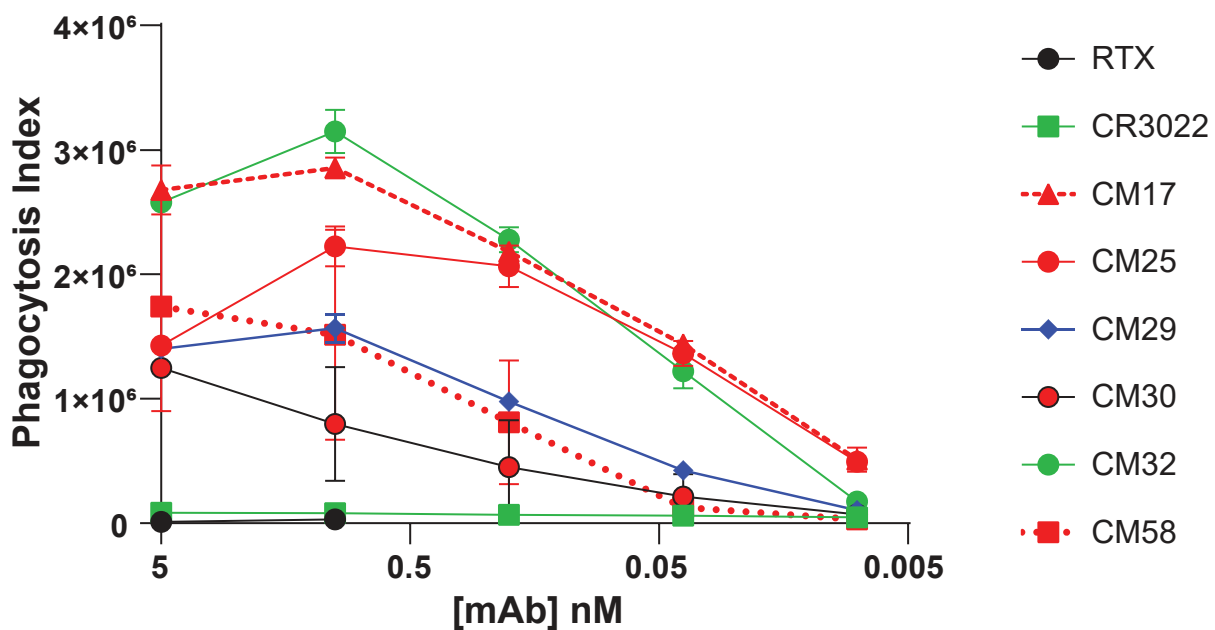


Fig. S4. ADCP ECD-bead assay using recombinant plasma IgG mAbs.

ADCP activity of recombinant plasma mAbs serially diluted on THP-1 cells in the presence of spike-conjugated, fluorescent polystyrene beads. The phagocytosis index metric represents the percent of bead-positive THP-1 cells multiplied by the average MFI of each cell to account for increased levels of bead internalization.

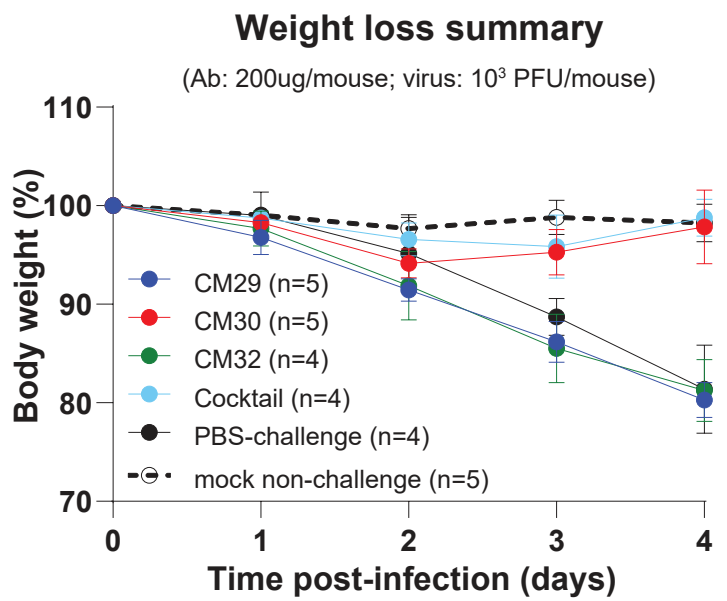
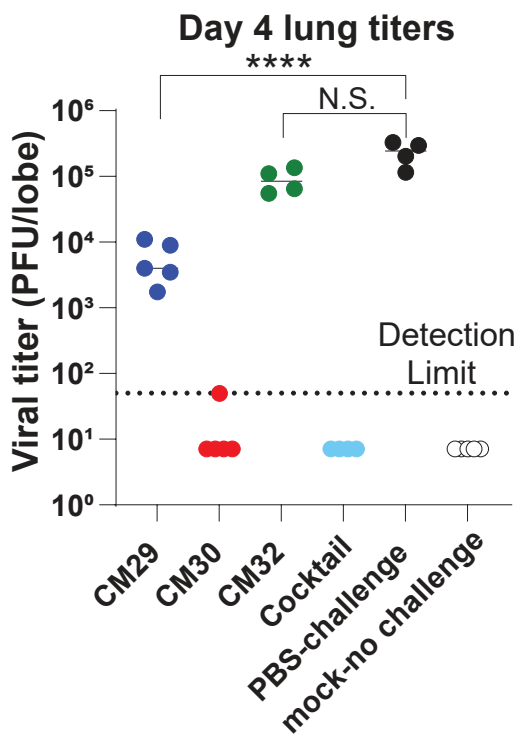


Fig. S5. In vivo protection against SARS-CoV-2 low dose viral challenge using recombinant plasma IgG mAbs.

Day 4 lung viral titers and average cohort weight loss of 12-month-old BALB/C mice after intranasal challenge with 10³ PFU (low dose) of mouse-adapted (MA10) SARS-CoV-2. A total of 200 µg mAb was administered 12 hours prior to challenge. Cocktail consisted of all four of the top donor P3 plasma IgG lineages (CM29, CM30, CM31, and CM32) in combination.

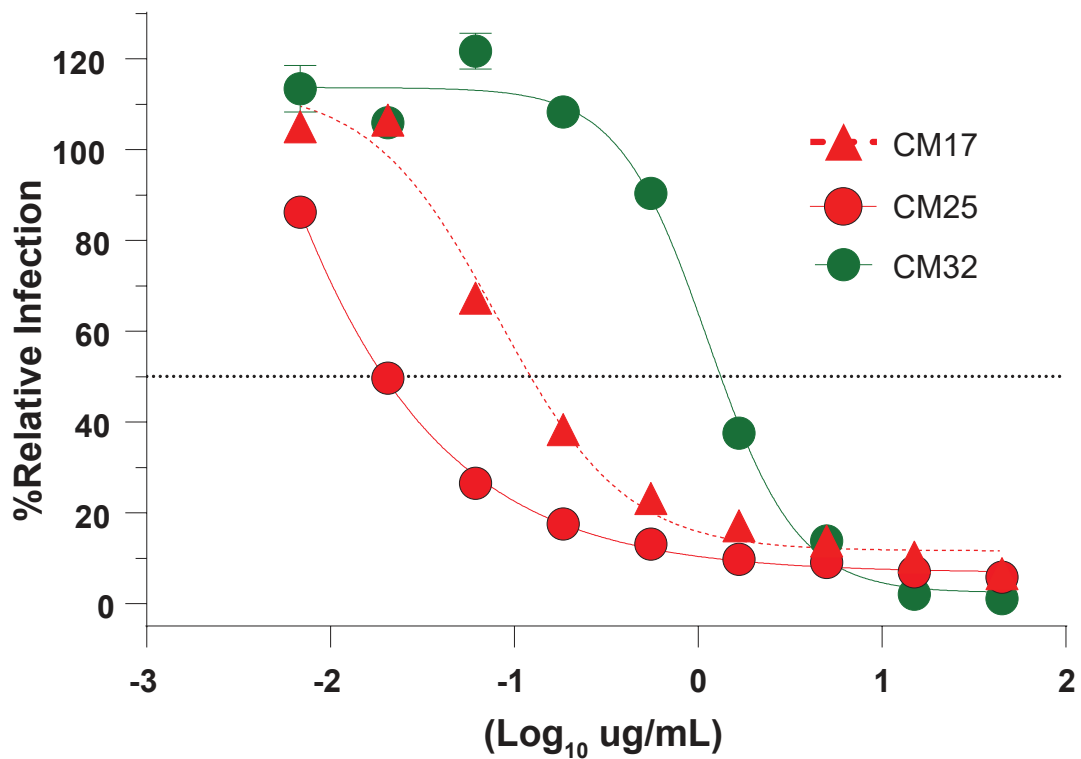


Fig. S6. Independent live virus neutralization titers of recombinant plasma IgG mAbs CM17, CM25, and CM32.

In vitro live virus neutralization curves for CM17, CM25, and CM32 repeated in second independent laboratory demonstrate similar levels of inhibition (as compared to data in Figs. 1D and 2C) of live SARS-CoV-2 virus infection of monolayered Vero E6 cells. The percent of infected Vero E6 cells in each sample dilution was normalized relative to the virus-only (no plasma) negative control sample.

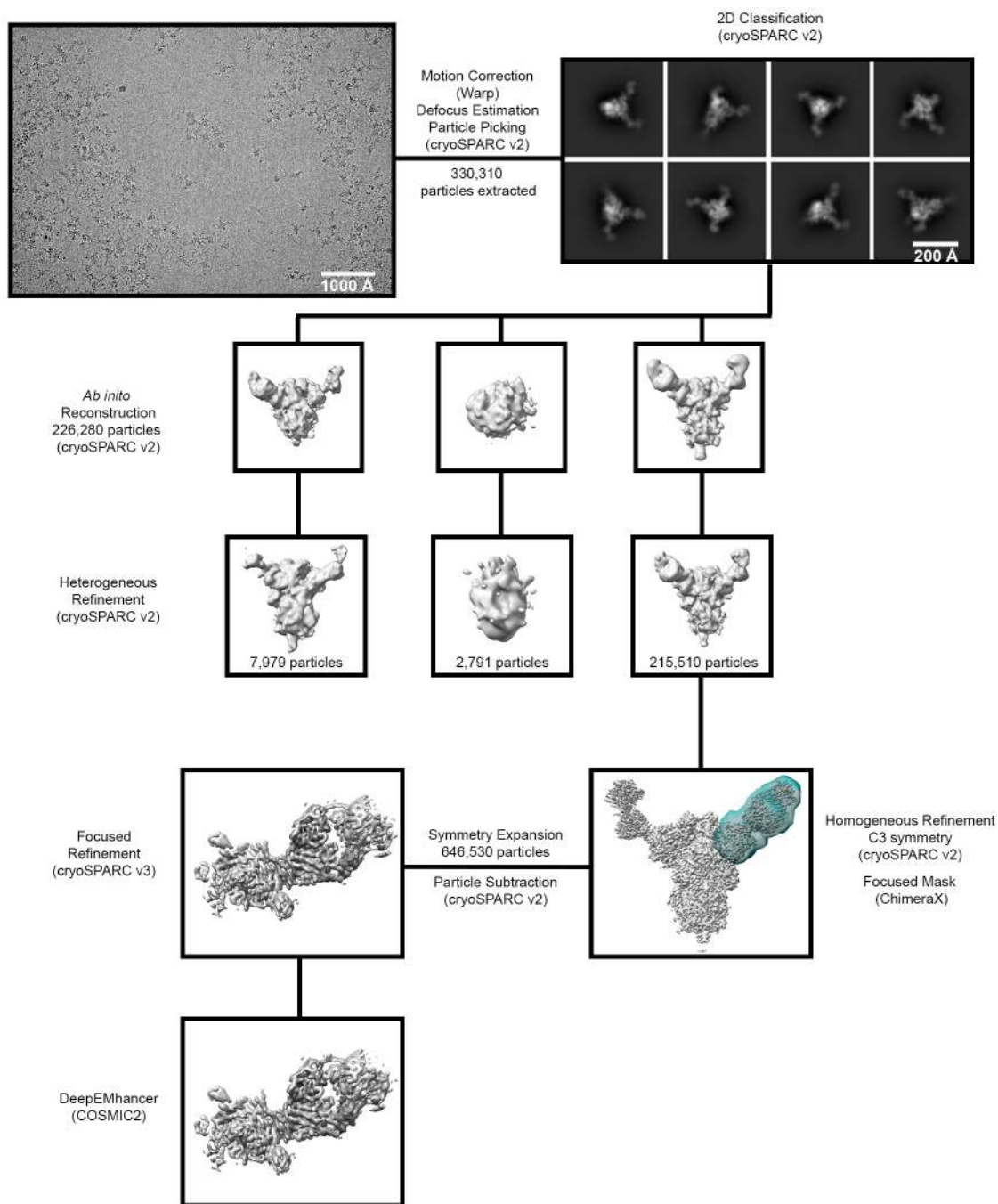


Fig. S7. Cryo-EM data processing workflow. Flowchart outlining cryo-EM data processing of SARS-CoV-2 S-ECD bound to CM25 Fab. Additional details can be found in the Methods section under "Cryogenic electron microscopy (cryo-EM)".

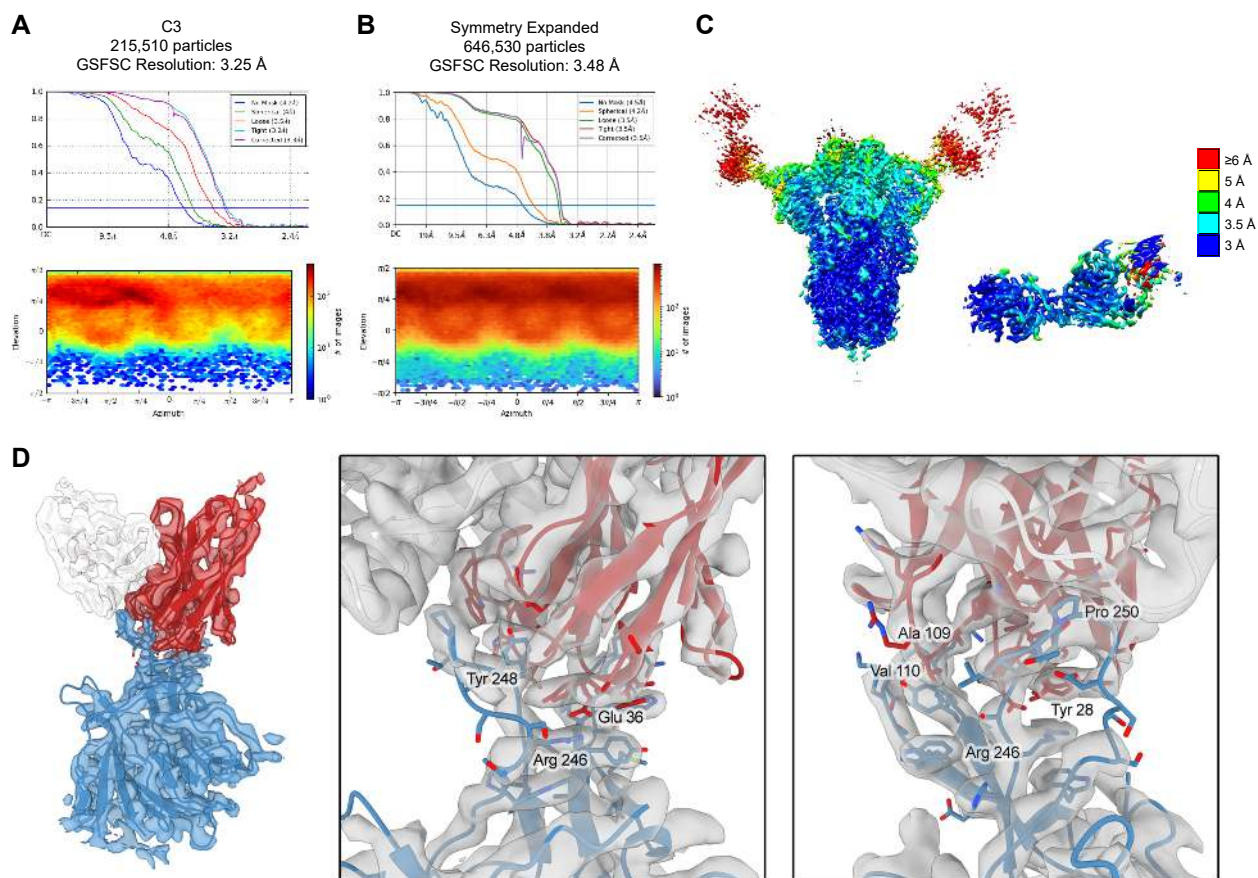


Fig. S8. Cryo-EM structure validation.

(A) FSC curve and viewing distribution plot for the overall S-CM25 Fab structure, generated in cryoSPARC v2.15. (B) FSC curve and viewing distribution plot for the focused refinement of S-NTD bound to the CM25 Fab. (C) Cryo-EM density of the overall S-CM25 (left) and the focused S-NTD-CM25 (right) reconstructions, colored according to local resolution. (D) Focused refinement density and corresponding models for the NTD (blue), CM25 Heavy chain (red), and CM25 Light chain (white). Full view of the NTD-CM25 Fab complex (left). Detailed views of the binding interface (middle, right). Oxygen atoms are colored red, nitrogen blue, and sulfurs yellow.

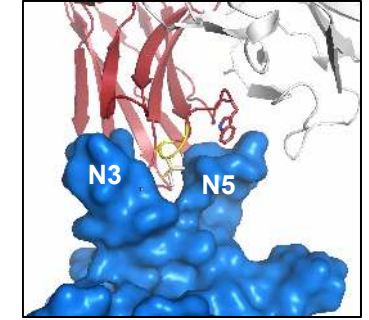
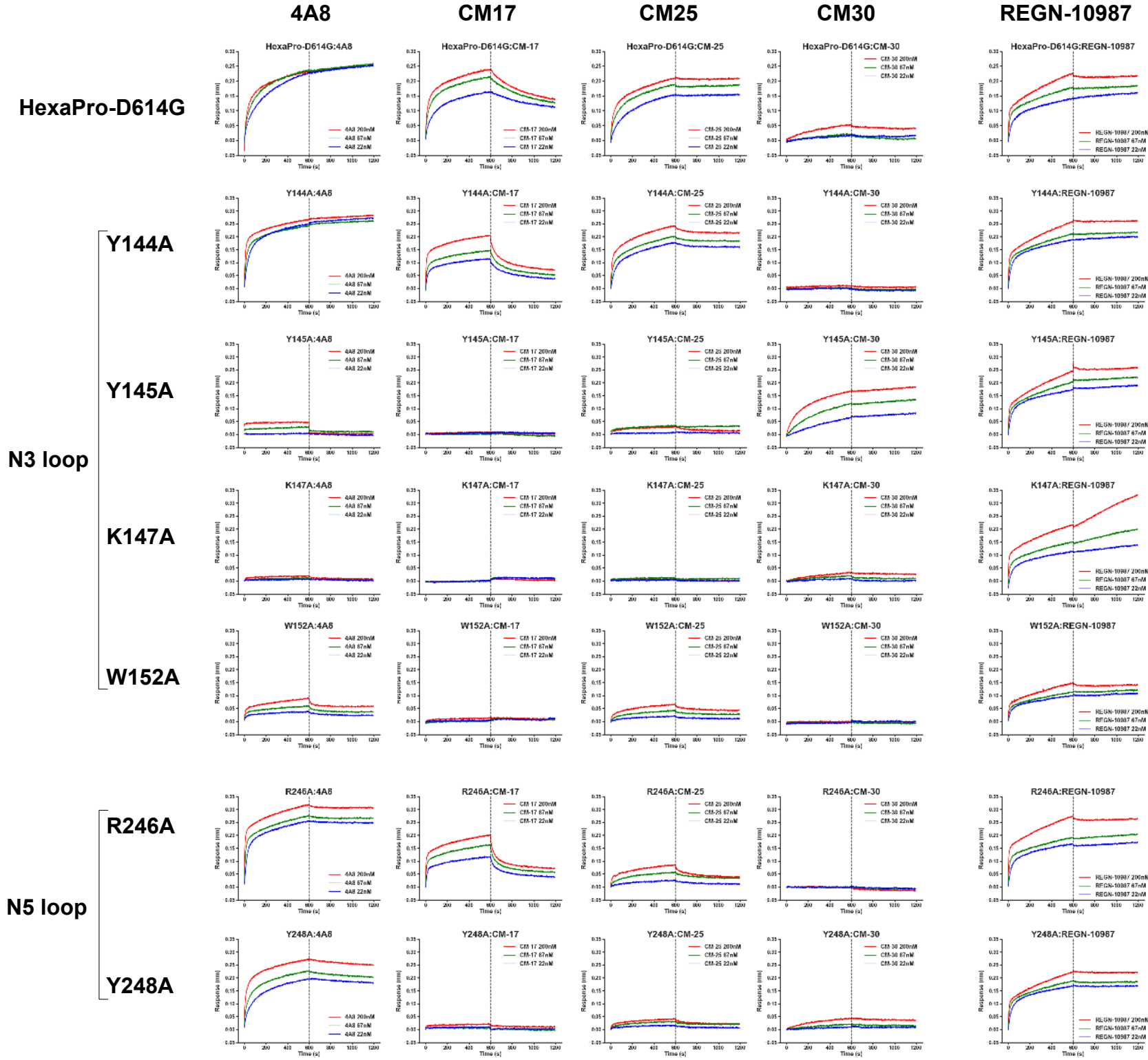


Fig. S9. Binding of anti-NTD NABs to SARS-CoV-2 spike (S-ECD) with mutations in the N3 and N5 loops. Biolayer interferometry (BLI) sensorgrams of anti-NTD IgG and control anti-RBD IgG REGN-10987. Four residues (Y144, Y145, K147, and W152) are located in the N3 loop of the S-ECD NTD and two residues (R246 and Y248) are located in the N5 loop of the S-ECD NTD.

Table S1: Cohort Information

Subject	Age	Sex	Plasma Neut. ID50 (95% CI) (early conv.)	Plasma Neut. ID50 (95% CI) (late conv.)	mAbs	Plasma IgG Lineages (early convalescence)		
						Total	RBD	Non- RBD
P1	65	F	12.5 - 16.6	14.8 - 21.1		22	4	18
P2	35	F	60.7 - 74.5	33.6 - 43.8	CM17 CM25	20	5	15
P3	60	M	7.4 - 10.2	n.d.	CM29 CM30 CM31 CM32	6	1	5
P4	47	M	248 - 326	127 - 169	CM58	17	4	13

Table S2: Recombinant plasma mAb binding and functionality

Ab	K _D (nM)	Subject	Specificity	IC ₅₀ (µg/mL)		Protection
	1:1 binding			indirect ELISA	authentic virus	MA10 mouse model
CM17	9.34±0.185	P2	NTD	0.031		Full (negligible weight loss, lung viral replication)
CM25	9.44±0.055	P2	NTD	0.012		Full (negligible weight loss, lung viral replication)
CM29	6.58±0.62	P3	S2	>20		Partial (-1.5 log ₁₀ reduction lung titer)
CM30	0.841±0.839	P3	NTD	0.83		Full (negligible weight loss, lung viral replication)
CM31	37.7±1.65	P3	NTD	>20		None
CM32	6±0.115	P3	RBD	2.1		None
CM58	13.9±0.02	P4	NTD	0.81		Partial (-2.5 log ₁₀ reduction lung titer)
S309	0.015	n/a	RBD	0.075		n.d.
CR3022	34.8±13.0	n/a	RBD	>20		n.d

Table S3: Convergent NTD mAbs

Ab	IGHV	IGHD	IGHJ	CDR-H3	AA	Light Chain
CM17	1-24	3-10	4	ATAAA <u>V</u> RGRGTIDY	14	IGLV1-51
CM25	1-24	3-10	5	ATGPA <u>V</u> RRGSWFDP	14	IGLV1-51
CM58	1-24	3-10	5	ATGP <u>V</u> RGVIGWFDP	14	IGLV2-14
COV2-2199 (8)	1-24	3-10	4	ATGF <u>A</u> VFGRAAVPY	14	IGLV3-1
COVA2-37 (7)	1-24	2-2	5	ATSPA <u>V</u> MSVGWVDP	14	IGLV1-40
4-A8 (4)	1-24	6-19	6	ATSTA <u>V</u> AGTPDLFDYYYYGMDV	21	IGKV2-24
1-68 (5)	1-24	6-19	6	ATGW <u>A</u> VAGSSDVWYYYYGMDV	21	IGLV2-18
1-87 (5)	1-24	6-19	6	ATGIA <u>V</u> IGPPPSTYYYYGMDV	21	IGLV2-14

Table S4: mAb–NTD interface analysis

4A8			CM25			NTD (4A8)			NTD (CM25)			
	ASA(Å²)	BSA(Å²)		ASA(Å²)	BSA(Å²)		ASA(Å²)	BSA(Å²)		ASA(Å²)	BSA(Å²)	
GLY27	63	10	CDR-H1	79	15	TYR144	57	32	N3 Loop	94	26	
TYR28	80	58		103	80	TYR145	85	81		103	75	
THR29	53	25		66	14	HIS146	71	25		75	22	
LEU30	4	3		25	11	LYS147	156	147		172	134	
THR35	106	80		112	80	ASN148	91	19		120	24	
GLU36	79	79		89	80	ASN149	116	0		83	0	
LEU37	13	12		31	29	LYS150	155	69		154	14	
						SER151	49	2		24	0	
PHE56	55	37	CDR-H2	62	40	TRP152	127	85	N5 Loop	148	5	
ASP57	46	0		23	0							
PRO58	25	13		37	3	HIS245	79	36		0	0	
GLU59	80	8		81	0	ARG246	85	72		85	62	
ASP62	127	11		100	4	SER247	46	12		64	16	
GLY63	37	24		53	30	TYR248	122	107		170	150	
						LEU249	182	90		153	54	
GLU80	78	10	CDR-H3	0	0	THR250	93	18		77	5	
						PRO251	190	0		125	32	
THR108	48	33		38	25	PRO108	GLY252	0		0	82	0
ALA109	26	24		32	22	ALA109	ASP253	0		0	66	1
VAL110	67	65		133	90	VAL110	SER254	0		0	65	0
ALA111	39	23		139	9	ARG111	SER255	0		0	89	29
GLY111.1	73	62		137	0	ARG112.1	SER256	125		6	0	0
THR111.2	74	30		49	0	GLY112	GLY257	30		18	0	0
PRO111.3	131	79		113	0	SER113						
ASP111.4	119	0		153	91	TRP114						
LEU112.4	124	0		98	1	PHE115						
PHE112.3	148	39		24	18	ASP116						
ASP112.2	89	0	73	2	PRO117							
TYR112.1	105	30										
TYR112	83	19										
TYR113	137	0										
GLY114	30	0										
MET115	62	0										
ASP116	72	11										

* bold indicates residues >50% buried surface area (BSA)

Table S5

EM data collection		
Microscope	FEI Titan Krios	
Voltage (kV)	300	
Detector	Gatan K3	
Magnification (nominal)	22,500	
Pixel size (Å/pix)	1.1	
Exposure rate (e ⁻ /pix/sec)	8	
Frames per exposure	80	
Exposure (e ⁻ /Å ²)	80	
Defocus range (μm)	1.5-3.0	
Tilt angle (°)	30	
Micrographs collected	3,339	
Micrographs used	1,074	
Particles extracted (total)	330,310	
Automation software	SerialEM	
Sample	SARS-CoV-2 S + CM25 Fab	
3D reconstruction statistics		
	Overall	NTD-CM25 subcomplex
Particles	215,510	646,530 (symmetry expanded)
Symmetry	C3	C1
Map sharpening B-factor	-174.5	-165.1
Unmasked resolution at 0.5 FSC (Å)	6.65	4.16
Masked resolution at 0.5 FSC (Å)	4.22	3.78
Unmasked resolution at 0.143 FSC (Å)	3.64	3.52
Masked resolution at 0.143 FSC (Å)	3.25	3.48
Model refinement and validation statistics		
Refinement package	Phenix	
Refinement tool	Real-space refinement	
Refinement strategies	min global, local_grid_search, adp, ss restraints, rotamer restraints, ramachandran restraints	
Composition		
Amino acids	453	
RMSD bonds (Å)	0.006	
RMSD angles (°)	0.96	
Average B-factors		
Amino acids	93.27	
Ramachandran		
Favored (%)	93.23	
Allowed (%)	6.77	

Outliers (%)	0
Rotamer outliers (%)	0
Clash score	11.44
C-beta outliers (%)	0
CaBLAM outliers (%)	2.54
CC (mask)	0.80
MolProbity score	2.01
EMRinger score	2.61

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